

Using highD and inD: a plan for the comfort-zone work and for assessing the published model

2026-09-17. Written after Jonas requested access to both datasets. A plan, not a result: nothing here has touched the data, and every count, rate and feasibility statement below is to be verified on download. Cards are numbered NC (comfort-zone boundaries) and NM (the published model); none is authorized until Jonas says so.

0 The short version

The two datasets can do more than docs/data_requirements.md §9 allowed them in August ("good for distributional calibration only; no driver state and no onsets"). That verdict was right for the question then asked, a per-driver onset label. Since then the model became gate $\times$ axis $\times$ level with a population percentile as the deliverable, and three things changed what drone data can offer:

- 1 The left turn needs no onset.** A turning driver's decision is binary and observable: the gap was accepted or rejected. inD gives that directly, and it is the same kind of response as the video study's "intervene or not" and the test track's Go/No-Go.
- 2 The deliverable is a population percentile, not a per-driver level.** Neither dataset identifies drivers across recordings, so per-driver levels and the trait (TR.1) are out of reach; the population level is not.
- 3 Two of the project's open questions are naturalistic by nature.** Card EX.2 made first exposure the primary setting, which predicts which fitted level should match unrehearsed drivers. And the method review's finding that the released model is unstable in plain car following (§4.2, cards GZ.1 and GZ.2) can only be judged against how real drivers follow.

So the plan has two programs sharing one ingest step. The comfort-zone program's strongest card is the inD left turn; the paper-assessment program's strongest card is sustained following on highD.

1 What each dataset gives, and what it does not

	highD	inD
Setting	German highways, drone, same-direction traffic only	four German urban intersections, drone
Road users (website)	about 110 500 vehicles, cars and trucks	about 8 200 vehicles and 5 300 pedestrians and cyclists
Per track	position, velocity, acceleration, size, lane, preceding and following ids, THW, TTC, lane-change annotation	position, heading, velocity, acceleration, bounding box, class
Maps	lane markings	OpenDRIVE and Lanelet2 maps of each intersection
Driver identity across recordings	none	none
Brake lights, pedal, gaze	none	none
To verify on download	the position reference point (box corner or centre), driving direction per lane, frame rate, the observation window per vehicle	frame rate, whether each intersection is signalized, which turning relations have oncoming or priority traffic, how many decisions each turning relation yields

Consequences that shape every card:

- Responses are revealed behavior, not judgments.** A video participant says "I would intervene"; a highD follower either decelerates or not. Mapping one onto the other is an assumption to state, as card TT.1 stated "Go on the track = not intervening on video".
- An onset has to be derived from kinematics.** Deceleration onset from 25 Hz tracks is feasible but has a detection latency and a false-positive rate that must be measured before any timing claim.
- The observation window is short.** A highD vehicle is in view for the length of the recorded road segment only, so events whose run-up or response falls outside it are censored.
- The license:** non-commercial research, no redistribution, citation required. Raw data stays in external/ (untracked); only aggregate results enter the repository. No data or per-track derived data goes to anyone else, the authors of the published model included.

- **Volvo Cars** [Jonas, 2026-09-17, NAT.Q2]: no highD or inD data goes to Volvo Cars; code and aggregate results may be shipped. The transfer policy already never bundles external/** from Chalmers. To keep per-track derived data out of reach of a bundle, every file holding one row per track, event or frame derived from highD or inD is written under external/highD_derived/ or external/inD_derived/, never under replication/; the reports and aggregate tables in replication/ carry no track ids. The loaders' property tests check that no report written by a naturalistic card contains a trackId column.

2 Step zero, shared by both programs

NC.0 — ingest, conventions and the jitter floor (a few days; gate for everything else). Loaders `src/comfortzone/highd.py` and `ind.py` with property tests that check the claims (a track's integrated velocity reproduces its positions; THW recomputed from positions matches the provided THW; lane changes counted from lane ids match the annotation). Establish the conventions from the data, not from documentation, as the overtake and left-turn loaders did: the reference point, the driving direction, heading. Measure the position and velocity jitter floor on steady driving, per dataset, before any spread or window is chosen (the trap card HS.1 fell into). Decide the route of the adapter (query NAT.Q2).

NC.0b — a validated deceleration-onset detector. [Revised 2026-09-17 on Jonas's ruling, NAT.Q1: "an acceleration threshold (with some duration, maybe even change in speed)".] A response is a deceleration episode that meets three conditions, each a parameter pre-registered in NC.0b:

parameter	meaning	grid for the sensitivity sweep	how the primary value is chosen
a_th	the follower's longitudinal acceleration falls below -a_th	0.5, 1.0, 1.5, 2.0 m/s ²	the smallest value whose false-positive rate on steady following is under 5% at the chosen duration, given the measured acceleration noise floor
T_min	it stays below -a_th for at least this long	0.3, 0.5, 1.0 s	the shortest duration that keeps the false-positive rate under 5%; not shorter than three frames
Δv_min	the speed drops by at least this much over the episode	none, 0.5, 1.0 m/s	none as primary unless the first two conditions alone pass throttle-release coasting; the sweep reports the effect

The onset time is the first frame of the qualifying episode at which the acceleration crosses -a_th. An episode already under way when the event starts is not a response to that event (censored, and counted). Validated on synthetic traces with known onsets plus the measured jitter, reporting the detection latency (crossing time minus the true start of deceleration) and the false-positive rate. The primary thresholds are fixed in NC.0b's docstring before any event is scored, and every response-dependent card reports the sweep beside its primary result. Stop rule: if no setting in the grid keeps the false-positive rate under 5% with a median latency under 0.5 s, no timing claim is made from highD, and the cards below fall back to "responded within the window, yes or no". The inD gap-acceptance cards need no onset.

NC.0c — the census. Counts, before any hypothesis is tested: highD cut-ins by gap and relative speed, steady-following episodes by speed and THW, lead-deceleration events by magnitude; inD turning decisions by relation and conflicting road-user class. Each card below has a minimum count, stated in its pre-registration; a card whose count is not met is not run.

3 The comfort-zone program

card	question	data	response	why it matters	cost
NC.1	Do real drivers accepting a left turn across oncoming traffic show the boundary the video and the track found?	inD	gap accepted or rejected	a third anchor after video (2.42 s at first exposure, 2.18 pooled) and track (2.45 s), with no onset needed	days
NC.2	Is distance, not time, the left-turn axis when speed varies naturally?	inD	same	card B.3.v2 chose distance on a design with one speed per cell; naturalistic speeds break that confound	hours, after NC.1
NC.3	Does the video-fitted cut-in level predict when real followers respond?	highD	deceleration onset or response yes/no	the out-of-sample test of gate × looming × level on real traffic; and a direct test of EX.2's choice	days

card	question	data	response	why it matters	cost
NC.4	How far ahead do real drivers anticipate, per scenario?	both	onset relative to the lane change's lateral onset; the turning driver's slowing relative to the oncoming arrival	answers PC1.Q4 empirically: the scenario-specific anticipation horizon	days
NC.5	Where does free car following sit relative to the comfort-zone field's boundary?	highD	chosen THW by speed	the static field's closed-form boundary (README table) against revealed behavior; shares its computation with NM.2	hours
NC.6	Do real cut-ins fall inside the video study's design space?	highD	none, descriptive	tells whether study 2's stimuli are representative, and what share of real cut-ins cross each percentile	hours, part of NC.0c

NC.1 in more detail, since it is the recommended first test. For each left turn with an oncoming or priority road user, the decision moment is when the turning vehicle commits or stops. At that moment read the conflicting road user's distance to the conflict point and its time to arrival, and whether the gap was accepted. Rejected gaps before an accepted one belong to the same driver, so the unit of resampling is the turning track. Fit the population psychometric on the distance axis (no per-driver term; there is one decision-maker per track). Pre-stated comparisons: the naturalistic median against the video's first-exposure and pooled medians and the track's, as differences with intervals; and which of the two video settings the naturalistic median is closer to, which is EX.2's prediction put to data. Two known gaps must be addressed in the pre-registration: the video and track boundaries are in post-encroachment time, which a gap-acceptance decision does not produce directly, so the comparison is made on distance at the decision moment (B.3.v2's axis); and inD's speeds are urban and may not cover the video's 50 km/h.

NC.3 in more detail. For each cut-in, compute the G.1 gate and the looming rate along the follower's view, exactly as the video pipeline does (`src/comfortzone/interface.py` already does this for the Volvo Cars interface). Predict, with the video-fitted population and nothing refitted, the probability that the follower responds within the window, and its timing. Score against chance and against a refit on highD, as TT.1 scored the video population on the track. Report the prediction at both EX.2 settings and pre-state that the first-exposure setting is expected to score better. Confounds to state in advance: the follower may already be decelerating, may change lane instead, and trucks follow at other gaps.

Deferred: the cross-scenario generalization test (fit on one scenario, predict another) needs the common scale of EL.2 and query EL.Q4; the per-driver trait is out of reach in both datasets.

4 The paper-assessment program

The published model was compared against a meta-analysis of SHRP2 and ANNEXT response times (rear-end), a UK simulator study (opposite-direction lateral incursion) and a Canadian simulator study (the held-out intersection). None of those is free naturalistic driving. The method review (`docs/method_review.md`) found two things that only naturalistic data can settle: the accumulator is far from zero in ordinary following (§4.2), and, since cards GZ.1 and GZ.2, the released configuration brakes by itself after about 2.6 s of steady following.

card	question	data	method	cost
NM.1	Does the released model hold real car following that real drivers hold?	highD steady-following episodes	the authors' model, unmodified, on replayed lead trajectories (the tier-2 adapter already used by GZ.1 and GZ.2), a stratified sample by speed and THW; rate of spontaneous re-plans and braking against the human follower in the same episode	overnight runs: 5 to 100 s of CPU per simulated step
NM.2	Is the model's calibrated following preference where real drivers follow?	highD following distributions	the preference function (<code>src/aidriver/preferences.py</code> , verified against the code) evaluated on real following states: the share of real following that the collision-and-safety term scores as a shortfall, which is what feeds the accumulator in NM.1	hours; same computation as NC.5

card	question	data	method	cost
NM.3	Does the response-time-to-urgency relation hold in naturalistic lead decelerations?	highD lead-deceleration events	onset from NC.0b, urgency as inverse tau at the lead's deceleration onset; the relation compared with the deposit's and with the paper's regression	days; stop rule if hard decelerations are too few
NM.4	Before the conflict, do priority-road drivers behave as the model's intersection scenario assumes?	inD straight-through drivers meeting vehicles entering from a minor road	approach speed and slowing when a vehicle waits or creeps, against the deposit's pre-event behavior in the intersection scenario	days; a feasibility check first, since failures to yield will be rare and the deposit's scenario geometry may not match any inD relation
—	the opposite-direction lateral incursion	not testable	highD has no opposing traffic; inD speeds are too low	—

What these cards can and cannot claim. They test the model on behavior it was not built to reproduce (sustained naturalistic following, free gap choice), which is a fair test of a model presented as a general account of driving but not a test of the paper's own comparisons. The write-up should say so in those words. The paper's own fit metrics (Jensen–Shannon for outcomes, Wasserstein for response-time distributions) can be applied where the naturalistic reference is a distribution of the same quantity, which is NM.3 and, for braking onsets, NM.1.

5 Order

[Revised 2026-09-17 on Jonas's priority, NAT.Q4: the comfort-zone work first, and of the paper cards NM.1.]

- 1 **On access:** NC.0, NC.0b, NC.0c. The census decides which of the cards below have the events they need.
- 2 **The comfort-zone core:** NC.1 and NC.2 (inD, no onset), then NC.3 and NC.4 (highD, onset from NC.0b). NC.6 comes free with the census.
- 3 **NM.1, started early because it is machine time:** pre-register it as soon as NC.0 has produced steady-following episodes, and run it overnight while the comfort-zone cards are worked on by day.
- 4 **Later:** NC.5, NM.2, NM.3; NM.4 only after a feasibility check.

Sharing [Jonas, 2026-09-17, NAT.Q3]: results are nominally for us; Jonas may share parts with Julian Schumann, Johan Engström and Arkady Zgonnikov. Reports are therefore written so that a section stands on its own, marks what is ours and unpublished, and never contains per-track data (which the license forbids sharing in any case). Nothing is sent by a session.

Each card follows the standing rules: a pre-registered script, a generated report, a worklog entry, queries, the suite green, one commit.

6 Decisions

Answered by Jonas on 2026-09-17:

- **NAT.Q1**, the response definition: an acceleration threshold with a minimum duration, possibly with a change in speed; parameterized and swept in NC.0b (§2). For inD, a rejected gap.
- **NAT.Q2**, the Volvo Cars pipeline: highD cut-ins may go through the split-site interface schema as a dry run; no highD or inD data goes to Volvo Cars, code and results may (§1).
- **NAT.Q3**, sharing: nominally internal; Jonas may share parts with Julian Schumann, Johan Engström and Arkady Zgonnikov (§5).
- **NAT.Q4**, priority: the comfort-zone cards first, and NM.1 of the paper cards (§5).

Still open, to settle in NC.0 or with Jonas when the card is written: the mapping assumption that "would intervene" on video corresponds to a highD response as defined in NC.0b is stated in every report that relies on it; whether a coasting follower (small deceleration, no threshold crossing) counts as having responded is exactly what the $\Delta v_{\min}$ sweep will show.

7 Where I may be wrong

- The dataset facts in §1 come from the dataset pages, not from the data. inD in particular may have few left turns with oncoming traffic, in which case NC.1 moves to turns yielding to priority traffic from the side, a different geometry from the video study.
- A deceleration onset may be a poor proxy for "uncomfortable": followers can release the throttle and coast, which drone kinematics show only as a small deceleration. NC.0b's stop rule protects the timing claims, not the mapping.
- NM.1 replays a recorded lead into a closed-loop model whose own speed diverges from the human follower's; comparing the two after divergence compares different situations. The report must compare the first spontaneous re-plan and braking onset, before divergence, not whole trajectories.
- The first-exposure prediction for NC.1 and NC.3 assumes naturalistic drivers correspond to the first-session state of a video participant. EX1.Q3 applies: session and exposure are confounded in study 1, so a better match to first exposure could also reflect session-day effects.